Jutsu Super

Time Limit	1 s
Memory Limit	128 MB

Pisi adalah legenda petarung yang populer di negeri sihir, suatu hari dia berkelana untuk mencari ilmu untuk mengupgrade kekuatannya. Pisi menemukan kuil sang master logika yang konon katanya menyimpan sebuah jurus yang disebut “Jutsu Super”, di tempat itu dia bertemu master logika dan meminta dia untuk mengajarinya Jutsu Super. Karena master logika adalah orang yang pelit ilmu, dia hanya mau memberikan ilmu yang ia punya kepada orang yang dapat memecahkan teka-tekinya, sehingga Pisi dibawa master logika ke ruang teka-teki. Di tempat itu Pisi melihat petak berukuran $\boldsymbol{N} \times \boldsymbol{M}$ dan masing-masing petak merupakan elemen yang disimbolkan oleh huruf **'A'** hingga **'Z'**. Sang master logika mendefinisikan kekuatan masing-masing petak sebagai hasil nilai minimum di antara panjang maksimum substring palindrom yang berpusat pada petak $P_{i,j}$, yang dibaca secara vertikal, horizontal, diagonal naik (/), dan diagonal turun(\).
Secara formal nilai kekuatan $P_{i,j}$ adalah sebagai berikut,

$$Kekuatan(P_{i,j}) = min\left(A(P_{i,j}), B(P_{i,j}), C(P_{i,j}), D(P_{i,j})\right)$$

Note* :

- $A(P_{i,j})$ = Panjang maksimum substring palindrom vertikal yang berpusat di $P_{i,j}$
- $B(P_{i,j})$ = Panjang maksimum substring palindrom horizontal yang berpusat di $P_{i,j}$
- $C(P_{i,j})$ = Panjang maksimum substring palindrom diagonal naik (/) yang berpusat di $P_{i,j}$
- $D(P_{i,j})$ = Panjang maksimum substring palindrom diagonal turun (\) yang berpusat di $P_{i,j}$

Karena berpusat di $P_{i,j}$, panjang setiap palindrom tersebut selalu ganjil dan minimal 1 (petak itu sendiri).

Master logika kemudian memberi Pisi tantangan, yakni untuk menghitung banyak kekuatan tertinggi di petak tersebut. Karena Pisi ingin sekali mendapat Jutsu Super tersebut, bantulah Pisi mendapatkannya.

Format Masukan

Baris pertama berisi 2 bilangan N dan M $(1 \leq N, M \leq 1000)$, yang menyatakan panjang baris dan kolom petak.

N baris berikutnya berisi M elemen $P_{i,j}$ tanpa spasi ($P_{i,j} \subseteq \{'A', 'B', 'C', \ldots, 'Z'\}$).

Format Keluaran

Dua buah bilangan bilangan bulat yang menyatakan banyak kekuatan tertinggi pada ruangan dan nilai kekuatan tertinggi.

Subsoal

1. (10 poin) Semua elemen sama
2. (20 poin) $N = 1$
3. (30 poin) $N, M \leq 50$
4. (40 poin) Tidak ada batasan tambahan

Contoh Masukan #1

```
5 5
CZLAA
NANAN
LXCXL
NANAN
CXLXC
```

Contoh Keluaran #1

```
3 3
```

Penjelasan untuk contoh masukan/keluaran #1

Petak dengan kekuatan tertinggi terletak pada $P_{3,3}$, $P_{4,2}$, $P_{4,4}$

Pada $P_{3,3}$, nilai kekuatannya yakni,

$Kekuatan(P_{3,3}) = min\left(A(P_{3,3}), B(P_{3,3}), C(P_{3,3}), D(P_{3,3})\right)$

$A(P_{3,3})$ = "LNCNL" = 5

$B(P_{3,3})$ = "LXCXL" = 5

$C(P_{3,3})$ = "ACA" = 3

$D(P_{3,3})$ = "CACAC" = 5

$Kekuatan(P_{3,3}) = min(5, 5, 3, 5) = 3$

$Kekuatan(P_{4,2}) = min(3,3,3,3) = 3$
$Kekuatan(P_{4,4}) = min(3,3,3,3) = 3$
Sehingga terdapat 3 petak dengan kekuatan tertinggi dengan nilai 3.

Contoh Masukan #2

```
3 10
ETETTOETTO
AYAYPKPBVB
ETETETEOTT
```

Contoh Keluaran #2

```
3 3
```

Contoh Masukan #3

```
3 4
ABCD
BCDA
CDAB
```

Contoh Keluaran #3

```
12 1
```

Contoh Masukan #4

```
10 10
ABCDEFGHIJ
BCDEFGHIJK
CDEFGLILKR
DEFGLIJKLM
EFOMLZLMNQ
FGHMJILMNO
GHMLKLMFOP
HXQIJMLIPQ
IJLJLNOPQR
JKQXNOPQRS
```

Contoh Keluaran #4

```
2 5
```

Jutsu Super (English Version)

Time Limit	1 s
Memory Limit	128 MB

Pisi is a legendary fighter who is famous in the land of magic. One day, Pisi travels in search of knowledge to upgrade his strength. Pisi finds the temple of the master of logic, which is said to hold a technique called the “Jutsu Super”. There, Pisi meets the master of logic and asks to be taught the Jutsu Super. The master is stingy with knowledge and only teaches those who can solve the master's puzzle, so Pisi is taken to the puzzle room. In that room, Pisi sees a grid of $\boldsymbol{N} \times \boldsymbol{M}$ cells, and each cell is an element symbolized by a letter from **'A'** to **'Z'**. The master defines the power of each cell as the minimum of the lengths of the longest palindromes centered at cell $P_{i,j}$, read vertically, horizontally, along the rising diagonal (/), and along the falling diagonal (\).
Formally, the power of cell $P_{i,j}$ is,

$$Power(P_{i,j}) = min\left(A(P_{i,j}), B(P_{i,j}), C(P_{i,j}), D(P_{i,j})\right)$$

Note* :

- $A(P_{i,j})$ = length of the longest vertical palindrome centered at $P_{i,j}$
- $B(P_{i,j})$ = length of the longest horizontal palindrome centered at $P_{i,j}$
- $C(P_{i,j})$ = length of the longest rising diagonal (/) palindrome centered $P_{i,j}$
- $D(P_{i,j})$ = length of the longest falling diagonal (\) palindrome centered at $P_{i,j}$

Since every palindrome above is centered at $P_{i,j}$, its length is always odd and at least 1 (the cell itself).

The Master of Logic then presented Pisi with a challenge,that is to calculate the number of instances of the highest power value on the cell. Eager to obtain the Jutsu Super, Help Pisi fulfill his dream.

Input Format

The first line contains two integers N and M ($1 \leq N, M \leq 1000$), the number of rows and columns of the grid.
Each of the next N lines contains M elements $P_{i,j}$ without space ($P_{i,j} \subseteq \{‘A’, ’B’, ‘C’, \dots, ‘\text{Z}’\}$).

Output Format

Output two integers, the number of cells with the highest power, and the value of the highest power.

Subtasks

1. (10 points) All elements are the same
2. (20 points) $N = 1$
3. (30 points) $N, M \leq 50$
4. (40 points) No additional constraints

Sample Input #1

```
5 5
CZLAA
NANAN
LXCXL
NANAN
CXLXC
```

Sample Output #1

```
3 3
```

Explanation for sample input/output #1

The cells with the highest power are $P_{3,3}$, $P_{4,2}$, $P_{4,4}$

At $P_{3,3}$, the power is,

$Power(P_{3,3}) = min\left(A(P_{3,3}), B(P_{3,3}), C(P_{3,3}), D(P_{3,3})\right)$

$A(P_{3,3})$ = "LNCNL" = 5

$B(P_{3,3})$ = "LXCXL" = 5

$C(P_{3,3})$ = "ACA" = 3

$D(P_{3,3})$ = "CACAC" = 5

$Power(P_{3,3}) = min(5,\ 5,\ 3,\ 5) = 3$

$Power(P_{4,2}) = min(3,\ 3,\ 3,\ 3) = 3$

$Power(P_{4,4}) = min(3,\ 3,\ 3,\ 3) = 3$

So there are 3 cells with the highest power, which is 3.

Sample Input #2

```
3 10
ETETTOETTO
AYAYPKPBVB
ETETETEOTT
```

Sample Output #2

```
3 3
```

Sample Input #3

```
3 4
ABCD
BCDA
CDAB
```

Sample Output #3

```
12 1
```

Sample Input #4

```
10 10
ABCDEFGHIJ
BCDEFGHIJK
CDEFGLILKR
DEFGLIJKLM
EFOMLZLMNQ
FGHMJILMNO
GHMLKLMFOP
HXQIJMLIPQ
IJLJLNOPQR
JKQXNOPQRS
```

Sample Output #4

```
2 5
```